

Universal Radiative Corrections for Precision Determinations of V_{ud} BEACH conference

Ana Costa Pereira, Martin Gorbahn, Francesco Moretti, Kai
Sieja, Emmanuele Stamou

Ana.Costa-Pereira@liverpool.ac.uk

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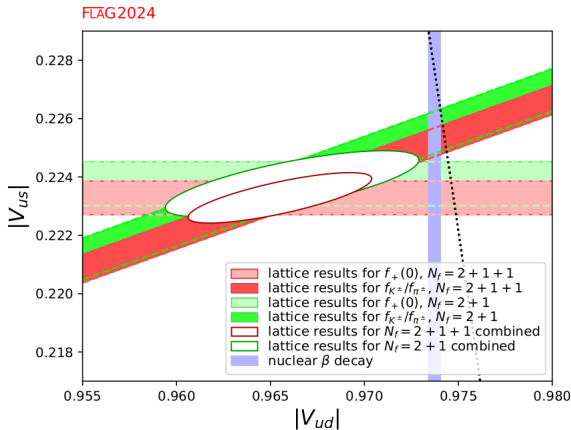
- 1 CKM matrix
- 2 V_{us} from kaon decays
- 3 V_{ud} EFT framework
- 4 V_{ud} from pion decay
- 5 V_{ud} from neutron decay
- 6 NLO EW corrections
- 7 NNLO EW corrections

- **CKM matrix** $\longrightarrow$ quark flavour mixing in weak interactions;
- First-row unitarity test

$$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1,$$

provides a sensitive probe of the SM;

- Current determinations of V_{ud} and V_{us} show a mild tension $\longrightarrow$ **Cabibbo Angle Anomaly**;
- Precision measurements of weak decays and accurate theoretical inputs are needed to clarify this tension;
- This talk:
 - Status of V_{us} from kaon decays ($K_{\ell 3}$ and $K_{\ell 2}$);
 - Status of V_{ud} from superallowed β decays, pion decay and neutron decay;
 - Ongoing work on **NNLO EW corrections for neutron decay** and V_{ud} extraction



V_{ud} and V_{us} using lattice QCD input and V_{ud} extracted from nuclear β decays, FLAG2024.

FLAG2024, Seng(2502.04721)

 K_{l3} decays:

$$K \rightarrow \pi \ell \nu_\ell, \quad \ell = e, \mu,$$

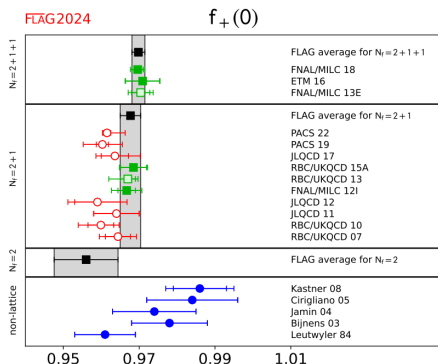
$$\Gamma(K_{l3}) = \frac{G_F^2 M_K^5}{192 \pi^3} C_K^2 S_{\text{EW}} |V_{us}|^2 |f_+^{K^0 \pi^-}(0)|^2 I_{K\ell}^{(0)} \left(1 + \delta_{SU(2)}^{K\pi} + \delta_{\text{EM}}^{K\ell}\right)^2$$

- $S_{\text{EW}} = 1.0232(3)$: process-independent electroweak correction;
- $I_{K\ell}^{(0)}$: phase-space integral in the isospin limit;
- $\delta_{SU(2)}^{K\pi}$: strong isospin-breaking correction;
- $\delta_{\text{EM}}^{K\ell}$: long-distance electromagnetic correction.

$$\langle \pi(p_\pi) | \bar{s} \gamma^\mu u | K(p_K) \rangle = (p_K + p_\pi)^\mu f_+(q^2) + (p_K - p_\pi)^\mu f_-(q^2),$$

Experimental input: $|V_{us}| f_+(0)$ **Hadronic input:** $f_+(0)$ from the vector form factor $\langle \pi | \bar{s} \gamma^\mu u | K \rangle$

Lattice input for $f_+(0)$



Lattice determinations of $f_+(0)$. Black squares and grey bands denote FLAG averages.

Experimental input:

$$|V_{us}|f_+(0) = 0.21654(41)$$

FLAG average:

$$f_+(0) = 0.9698(17)$$



$$|V_{us}| = 0.22328(58)$$

$$K_{\mu 2(\gamma)} : \quad K^+ \rightarrow \mu^+ \nu_\mu (\gamma)$$

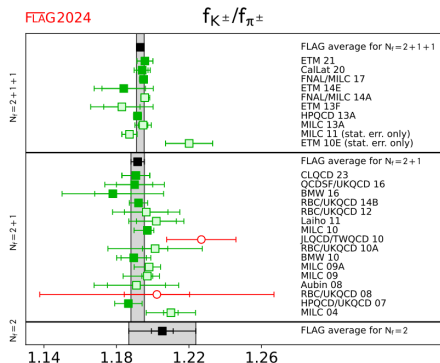
$$\frac{\Gamma(K_{\mu 2(\gamma)}^\pm)}{\Gamma(\pi_{\mu 2(\gamma)}^\pm)} = \left| \frac{V_{us}}{V_{ud}} \right|^2 \left(\frac{f_{K^\pm}}{f_{\pi^\pm}} \right)^2 \frac{M_{K^\pm} (1 - m_\mu^2/M_{K^\pm}^2)^2}{M_{\pi^\pm} (1 - m_\mu^2/M_{\pi^\pm}^2)^2} (1 + \delta_{\text{EM}})$$

$$\langle 0 | \bar{q}_1 \gamma^\mu \gamma_5 q_2 | P(p) \rangle = i p^\mu f_P, \quad P = \pi^+, K^+.$$

- Hadronic input encoded in the decay-constant ratio $f_{K^\pm}/f_{\pi^\pm}$;
- δ_{EM} : electromagnetic radiative correction;
- Lattice-QCD isospin corrections are systematically improvable;
- Isospin-breaking effects are relevant at current CKM precision.

$$\left| \frac{V_{us}}{V_{ud}} \right| \frac{f_{K^\pm}}{f_{\pi^\pm}} = 0.27599(41) \quad (\text{experiment})$$

Lattice determination of $f_{K^\pm}/f_{\pi^\pm}$



Lattice determinations of $f_{K^\pm}/f_{\pi^\pm}$. Black squares and grey bands denote FLAG averages.

Experimental input:

$$\left| \frac{V_{us}}{V_{ud}} \right| \frac{f_{K^\pm}}{f_{\pi^\pm}} = 0.27599(41)$$

FLAG average:

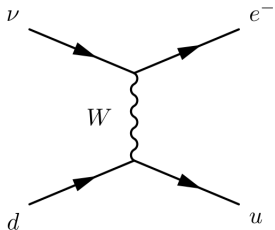
$$\frac{f_{K^\pm}}{f_{\pi^\pm}} = 1.1934(19)$$



$$\left| \frac{V_{us}}{V_{ud}} \right| = 0.23126(50)$$

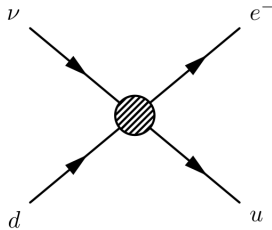
$(K_{\ell 2}/\pi_{\ell 2})$

$$d \rightarrow u e \bar{\nu}_e$$



Standard Model

$$m_W \gg k$$



Effective theory

$$\mathcal{H}_{\text{eff}} = \frac{G_F}{\sqrt{2}} V_{ud} C_q(\mu) Q_q$$

Cirigliano, Hoferichter, Valori (2602.11253)

$$\Gamma_{\pi\beta} = \frac{G_F^2 |V_{ud}|^2 M_\pi^5}{64\pi^3} |g_V^\pi|^2 |f_+^\pi(0)|^2 I_{\pi\ell} (1 + \Delta I_{\pi\ell})$$

- $f_+^\pi(0)$: pion vector form factor.
- $I_{\pi\ell}$: phase-space integral.
- $\Delta_{\text{RC}}^{\pi\ell}$: radiative corrections.
- g_V^π : effective vector coupling containing the short-distance electroweak corrections.

$$\Delta_{\text{RC}}^{\pi\ell} = 0.03403(11)$$

$$V_{ud}^\pi = 0.97346(283)$$

- The Behrends–Sirlin–Ademollo–Gatto theorem implies

$$f_+^\pi(0) \simeq 1 - 7 \times 10^{-6},$$

$$\frac{1}{\tau_n} = \frac{G_F^2 |V_{ud}|^2}{2\pi^3} m_e^5 f (1 + 3g_A^2) (1 + \Delta_R), \quad (1)$$

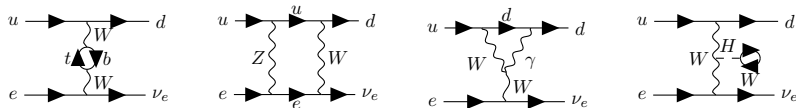
- f : phase-space factor.
- g_A : axial-vector coupling.
- Δ_R : electroweak radiative corrections.

$$\Delta_R^V = 2.436(16)\%,$$
$$|V_{ud}|_n = 0.97410(41).$$

- First NLL calculation of mixed $O(\alpha_s^2)$ corrections.
- Reduces theoretical uncertainty in Δ_R^V .
- Improves consistency of first-row CKM unitarity tests.

Matching condition

$$\mathcal{A}_{\text{SM}} = \mathcal{A}_{\text{EFT}} = V_{ud} C_q Q_q$$



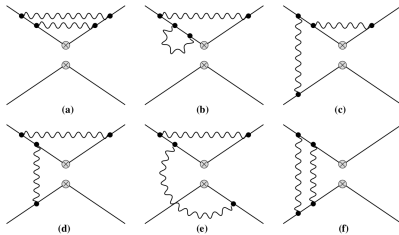
Representative one-loop EW diagrams contributing to the semi-leptonic process.

Wilson coefficient

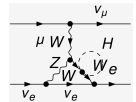
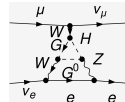
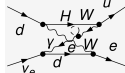
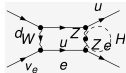
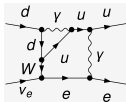
$$C_q(\mu) = 1 + \frac{\alpha}{4\pi} C_q^{\text{NLO}} + \left(\frac{\alpha}{4\pi}\right)^2 C_q^{\text{NNLO}}$$

$$C_q^{\text{NLO}} = -\frac{10}{3} - 2 \ln\left(\frac{\mu^2}{M_Z^2}\right)$$

- Matching performed at the electroweak scale.
- Universal corrections cancel in the ratio of the quark decay normalized to muon decay: $\frac{C_q}{C_\mu}$.
- Only box-diagram contributions survive.



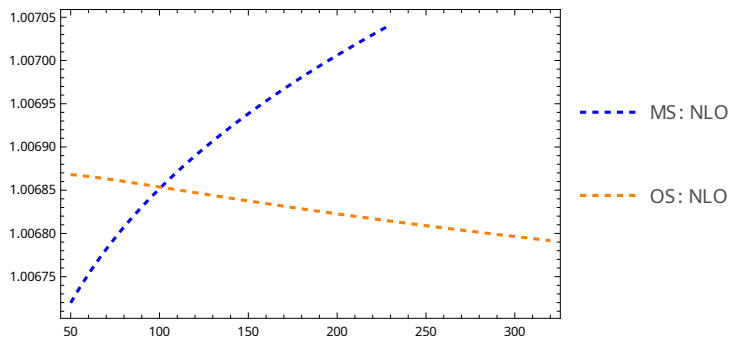
EFT NNLO EW corrections.



SM NNLO EW corrections.

- NNLO requires inclusion of boxes and penguins
- Tadpoles are included in the matching: gauge independent NNLO EW corrections and potentially substantial contribution in scheme conversion: $\overline{MS}$ to On-Shell.

Wilson-Coefficient Evolution



Scale dependence of the Wilson coefficient $C(2 \text{ GeV})$ at NLO.

- Residual scale dependence at NLO induces an uncertainty $\delta C(2 \text{ GeV}) \simeq \pm 2 \times 10^{-4}$, which is significant for precision determinations of V_{ud} .
- NNLO corrections reduce the uncertainty to $\delta C(2 \text{ GeV}) < 2 \times 10^{-5}$, making perturbative uncertainties essentially negligible.

- Combination of lattice inputs and perturbative calculations;
- Higher order calculation + sub-percent lattice calculations;
- Current ongoing work on NNLO EW corrections for V_{ud} extraction from neutron decay.